① a) $P(8) = C_8^8 p^8 q^0$

$= 1 \cdot (0.4)^8 \cdot (0.6)^0 = 0.00065536$

b) $P(3) = P(x \geq 3) - P(x \geq 2)$

$P(3) = 0.806 - 0.552$

$P(3) = 0.254$

c) $P(x \geq 4) = C_4^8 p^4 q^4$

$= 70 \cdot (0.4)^4 \cdot (0.6)^4$

$= 0.2322432$

(2)

a) $P(2) = \frac{\mu^2 e^{-\mu}}{2!} = \frac{(1.2)^2 \cdot e^{-1.2}}{2} = 0.2169$

$\mu = 1.2$

b) $P(x \leq 4) = P(0) + P(1) + P(2) + P(3) + P(4)$

$P(x \leq 4) = e^{-1.2}\left(\frac{1.2^0}{0!} + \frac{1.2^1}{1!} + \frac{(1.2)^2}{2!} + \frac{(1.2)^3}{3!} + \frac{(1.2)^4}{4!}\right)$

$P(x \leq 4) = 0.99225$

c) If avg. is 1.2 28.8 calls per 24 hours

If one hour get 8 calls 20.8 calls left for 23 hours

$\frac{20.8}{23} = 0.90435$ so it is not possible.